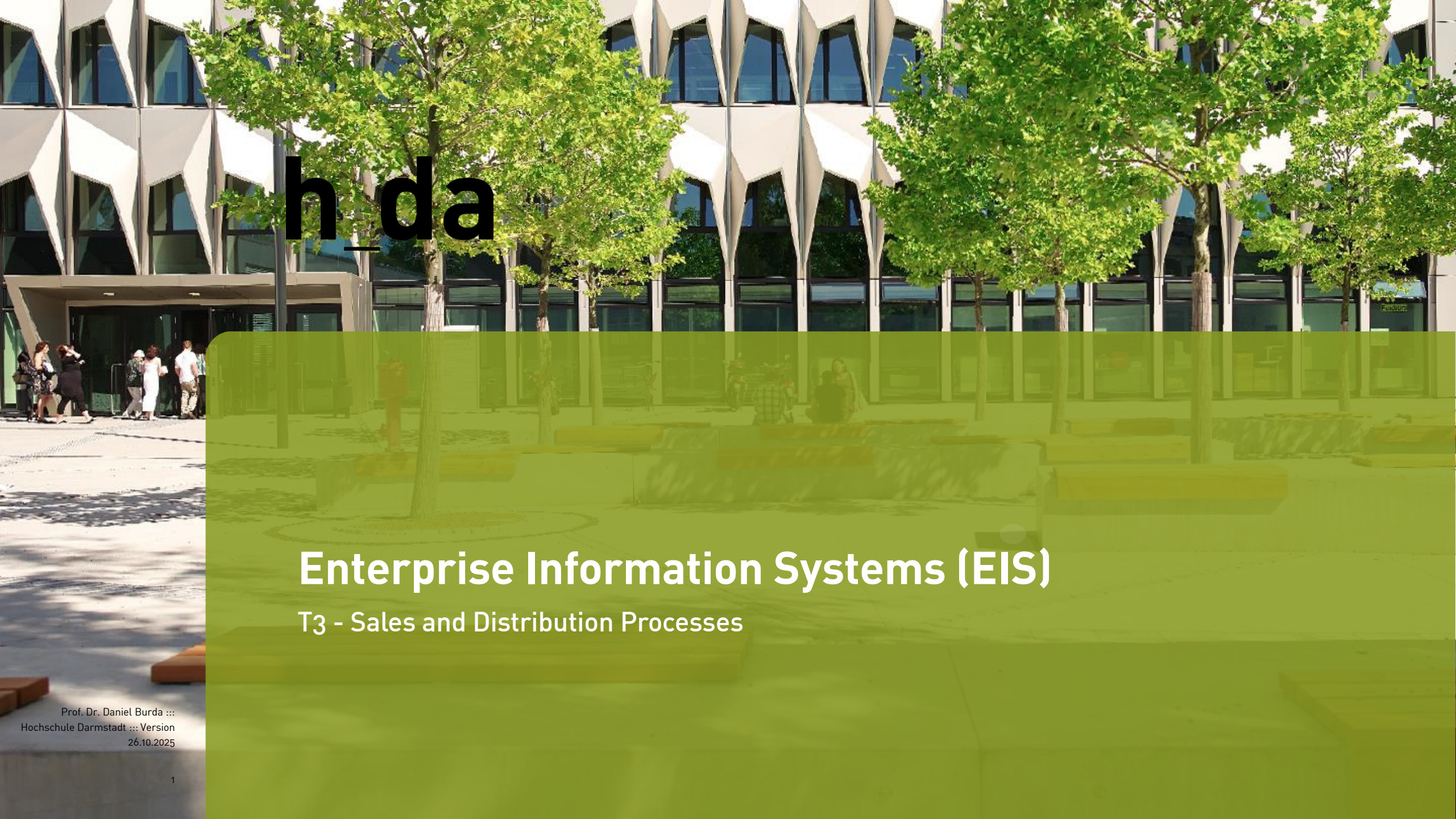


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Enterprise Information Systems (EIS)

T3 - Sales and Distribution Processes

Prof. Dr. Daniel Burda :::
Hochschule Darmstadt ::: Version
26.10.2025

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Schedule (When?)

Lecture				Practical			
No.	Date/Time/Room	Topic No.	Topic	PVL Relevant	Topic	Practical No.	Dates (2 Groups)
1	Tue., 07.10.2025, 10:15 - 11:45 D14 / 04.04	T0	Intoduction & Module Overview	---	---	---	---
2	Tue., 14.10.2025, 10:15 - 11:45 D14 / 04.04	T1	Enterprise Information Systems - Theoretical Perspective	---	---	---	---
3	Tue., 21.10.2025, 10:15 - 11:45 D14 / 04.04	T2	SAP Overview and Introduction S/4 HANA	---			
4	Tue., 28.10.2025, 10:15 - 11:45 D14 / 04.04	T3	Sales and Distribution Processes (1)	---	SAP Navigation Exercises	P#1	28.10.2025 / 29.10.2025
5	Tue., 04.11.2025, 10:15 - 11:45 D14 / 04.04	T3	Sales and Distribution Processes (2)	x	SD Case Study	P#2	04.11.2025 / 05.11.2025
6	Tue., 11.11.2025, 10:15 - 11:45 D14 / 04.04	T4	Materials Management and Procurement Processes (1)	x	SD Case Study	P#2	11.11.2025 / 12.11.2025
7	Tue., 18.11.2025, 10:15 - 11:45 D14 / 04.04	T4	Materials Management and Procurement Processes (2)	x	MM Case Study	P#3	18.11.2025 / 19.11.2025
8	Tue., 25.11.2025, 10:15 - 11:45 D14 / 04.04	T5	Financial Accounting Processes (1)	x	MM Case Study	P#3	25.11.2025 / 26.11.2025
9	Tue., 02.12.2025, 10:15 - 11:45 D14 / 04.04	T5	Financial Accounting Processes (2)	x	FI Case Study	P#4	02.12.2025 / 03.12.2025
10	Tue., 09.12.2025, 10:15 - 11:45 D14 / 04.04	T6	Integrated Project Management (1)	x	FI Case Study	P#4	09.12.2025 / 10.12.2025
11	Tue., 16.12.2025, 10:15 - 11:45 D14 / 04.04	T6	Integrated Project Management (2)	x	PS Case Study	P#5	16.12.2025 / 17.12.2025
12	Tue., 13.01.2026, 10:15 - 11:45 D14 / 04.04	T7	Human Capital Management Processes (1)	x	PS Case Study	P#5	13.01.2026 / 14.01.2026
13	Tue., 20.01.2026, 10:15 - 11:45 D14 / 04.04	T7	Human Capital Management Processes (2)	x	HCM Case Study	P#6	20.01.2026 / 21.01.2026
14	Tue., 27.01.2026, 10:15 - 11:45 D14 / 04.04	n/a	Test Exam / Q&A Final Exam	x	HCM Case Study	P#6	27.01.2026 / 28.01.2026

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Learning Objectives (What?)

Students can…

- … describe the **general order-to-cash process on the basis of the essential activities** and **differentiate it from an procurement process**.
- … describe the **organizational structures and master data** required for Sales and Distribution with SAP SD.
- … create **customer master data relevant for sales** with SAP SD.
- … **explain the integrated order-to-cash process and execute** it in SAP-SD

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Agenda

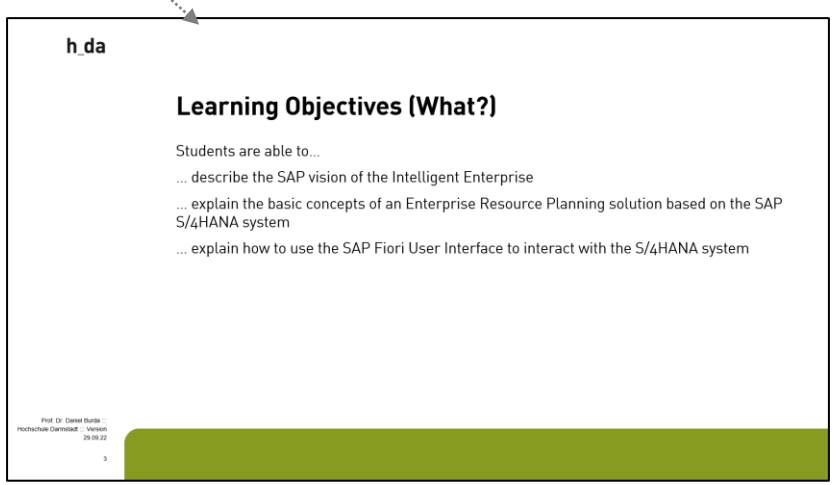
- Purchase-to-pay vs. Order-to-cash processes
- SD Organizational Structure
- SD Master Data
- Order-to-Cash Process
- Outlook SD Case Study

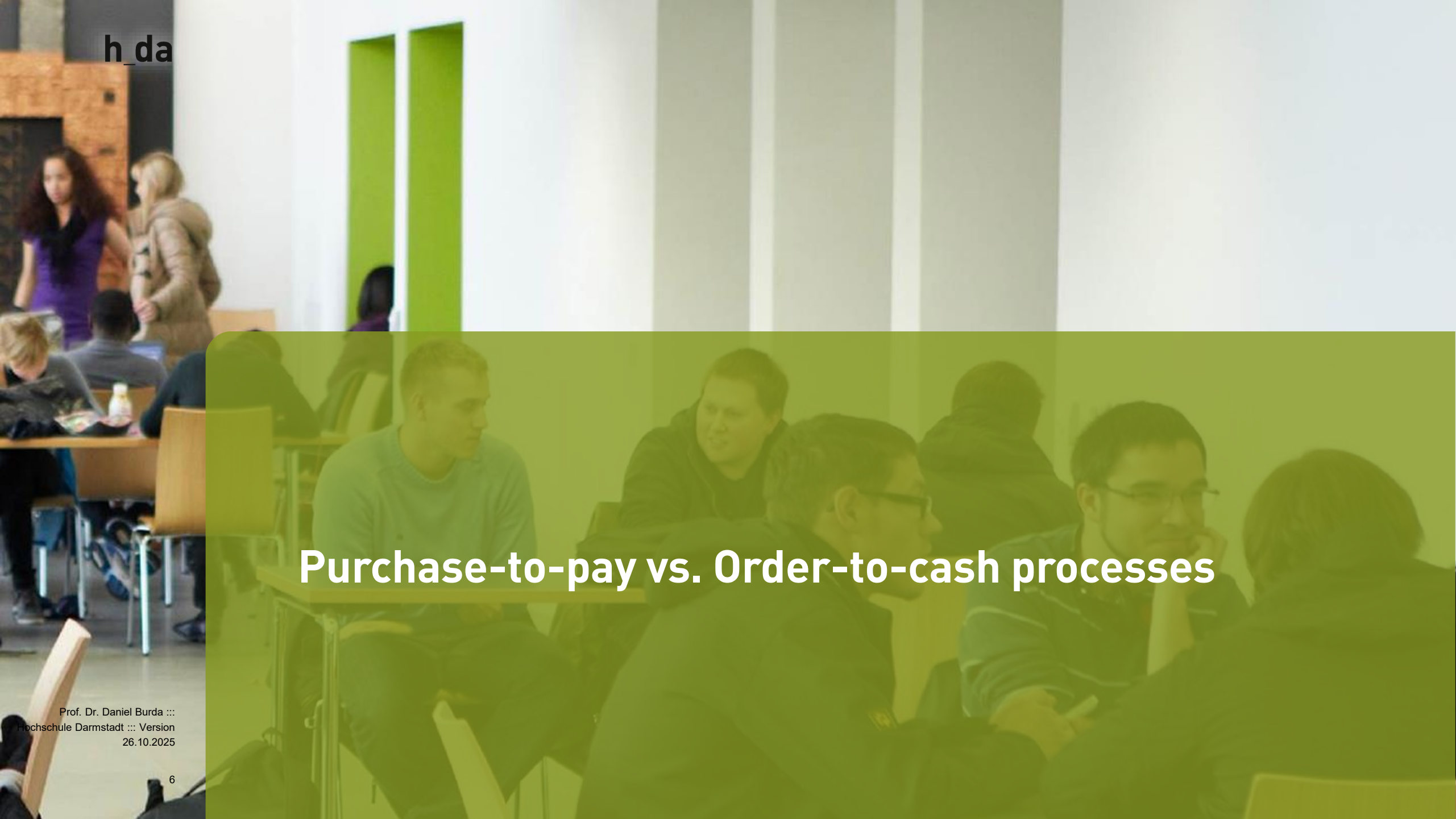
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Purchase-to-pay vs. Order-to-cash processes

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Purchase-to-pay vs. Order-to-cash

- "Purchase-to-pay/procure-to-pay" processes refer to all activities from the creation of an procurement order to payment of the due amounts to the supplier.
- This is the customer's view of a purchase order process and is therefore associated with the **procurement function** of the company.

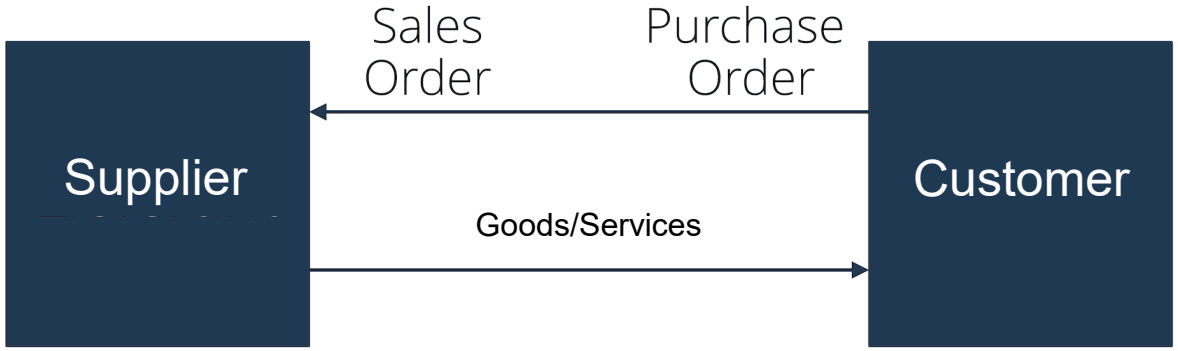
- "Order-to-cash" processes refer to all activities from the customer order to the receipt of payment.
- It is the supplier's view of an order process and is therefore associated with of the **sales function** in a company.



https://blog.esker.com/wp-content/uploads/2020/10/CCC_Petales_Cycle-US-1020x416.png

Purchase Order vs. Sales Order

- A purchase order is usually created/transmitted in the form of a document by a customer with a supplier.
- A purchase order sent by the customer becomes a sales order on the supplier side.
- A **purchase order is associated with the procurement** function in a company and is used in a purchase-to-pay process.
- A **sales order is associated with the sales function** of a company and is used in an order-to-cash process.



Quelle: https://ecosio.com/de/blog/unterschied-zwischen-einer-purchase-order-und-einer-sales-order/

Assignment

Assignment 1: Order-to-Cash Process

Tasks:

- Based on your understanding, model an order-to-cash process starting from the **first customer contact** until the customer **pays the invoice**.
- Concentrate on the essential process steps and on the **standard case ("happy path")** without exceptions, complications, etc.
- Present your process in the plenum.

Notes:

Choose a graphical notation of your choice for modeling, e.g. a simple flow chart or BPMN, EPK, UML.

You can use a modeling tool of your choice, e.g.: https://online.visual-paradigm.com/app/diagrams/#diagram:proj=0&type=Flowchart

Sales and Distribution (SD)

Curriculum: Introduction to S/4HANA using Global Bike



SAP University Alliances

SAP UCC Magdeburg

Teaching material - Information

Teaching material - Version

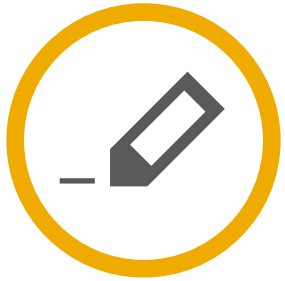
- 4.2 (July 2023)

- Software used
  - S/4HANA 2022
  - Fiori 3.0
- Model
  - Global Bike

- Prerequisites
  - No Prerequisites needed

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Module Information



Authors

- Bret Wagner
- Stefan Weidner
- Babett Ruß



Target Audience

- Beginner

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Unit Overview

- SD Organizational Structure
- SD Master Data
- Order-to-Cash Process

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Before processing Sales Orders an Organizational Structure has to be defined

Sales Organization

- An organizational unit responsible for the sale of certain products or services.
- The responsibility of a sales organization may include legal liability for products and customer claims

Distribution Channel

- The way in which products or services reach the customer
- Typical examples of distribution channels are wholesale, retail, or direct sales

Division

- A way of grouping materials, products, or services

Sales Area

- Combination of **Sales Organization, Distribution Channel and Division**
- Determines conditions (i.e. pricing) for sales activities

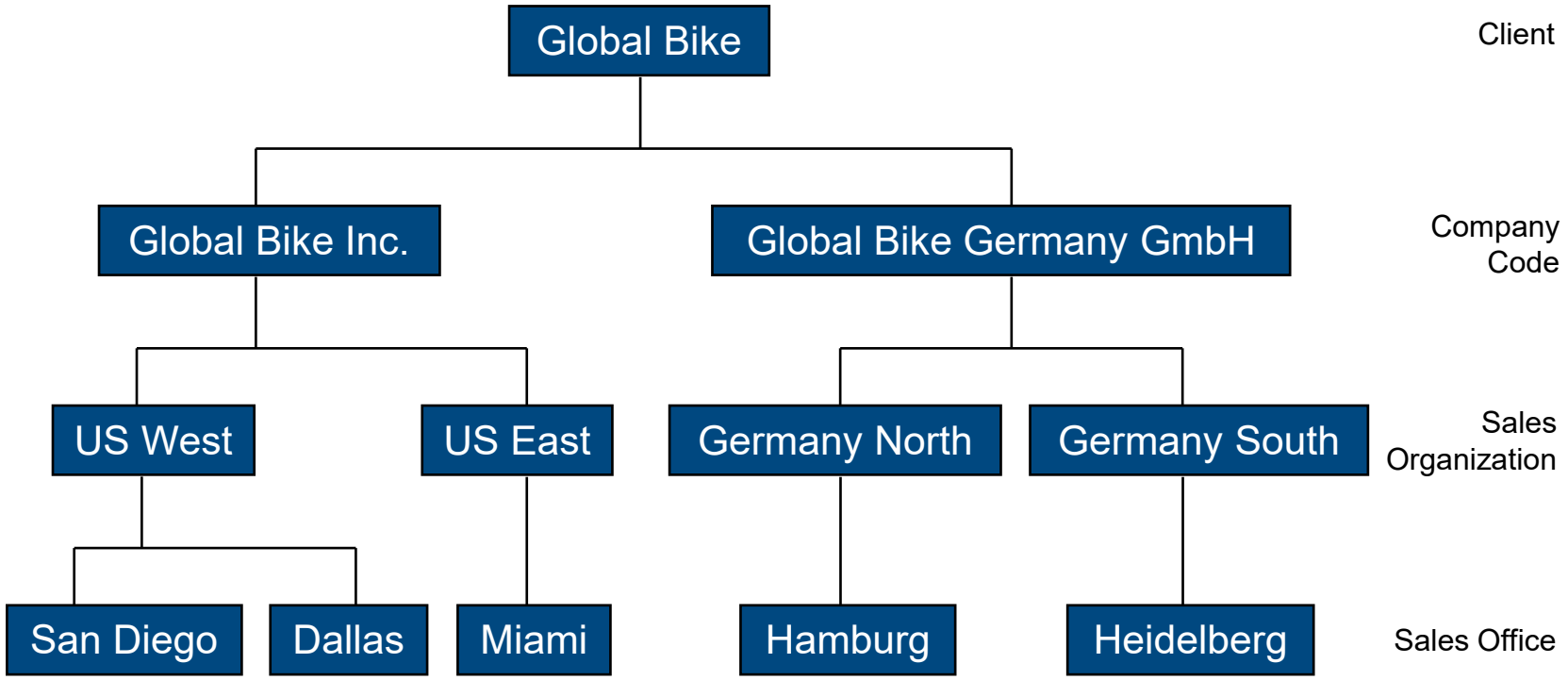
Plant (here: Delivering Plant)

- Plant from which the goods should be delivered to the customer

Other SD organization units:

- Shipping Point
- Loading Point
- Sales Office
- Sales Group
- Sales Person

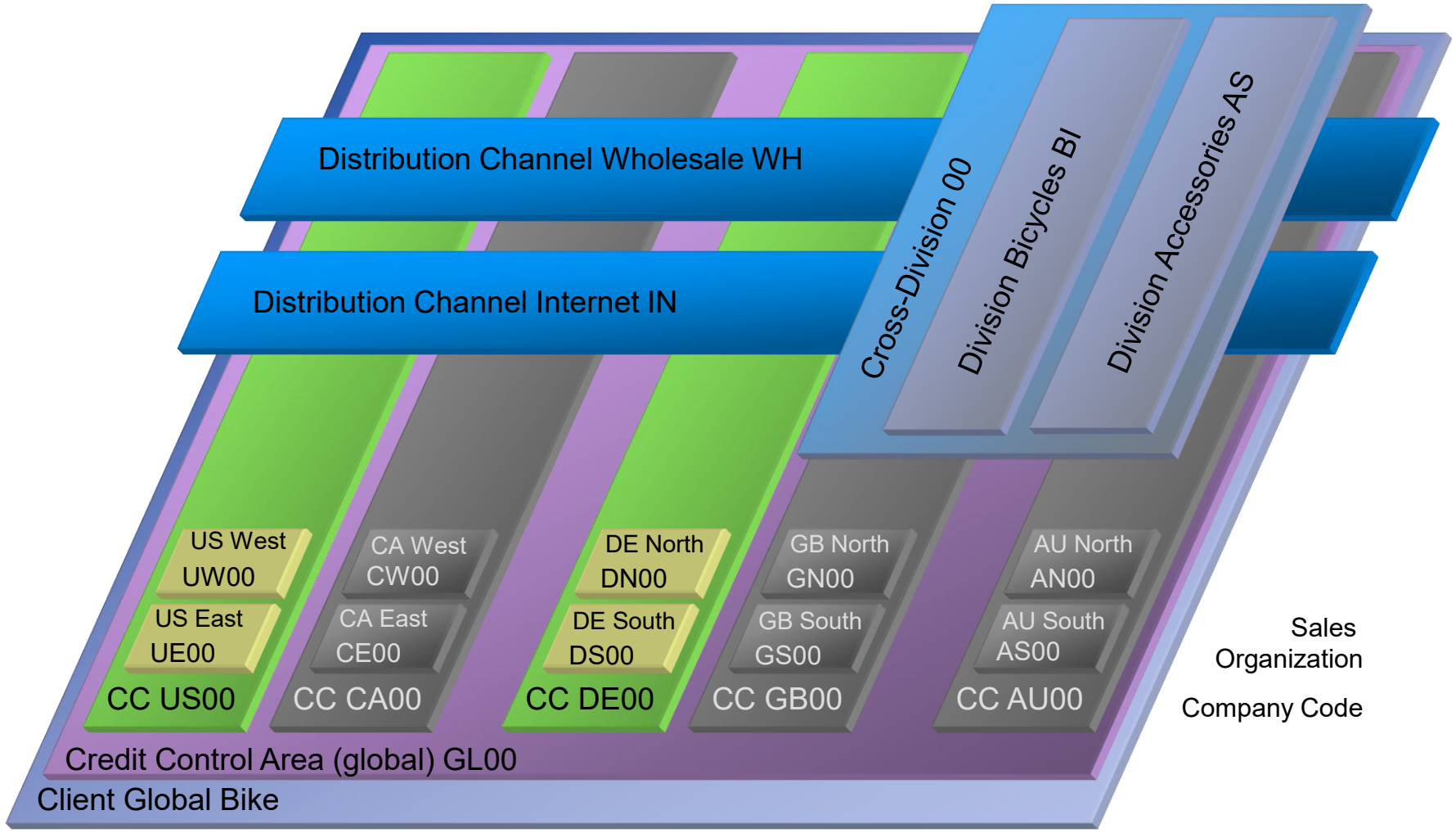
Global Bike Structure for Sales and Distribution



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Global Bike Enterprise Structure in SAP ERP (Sales)

Demo



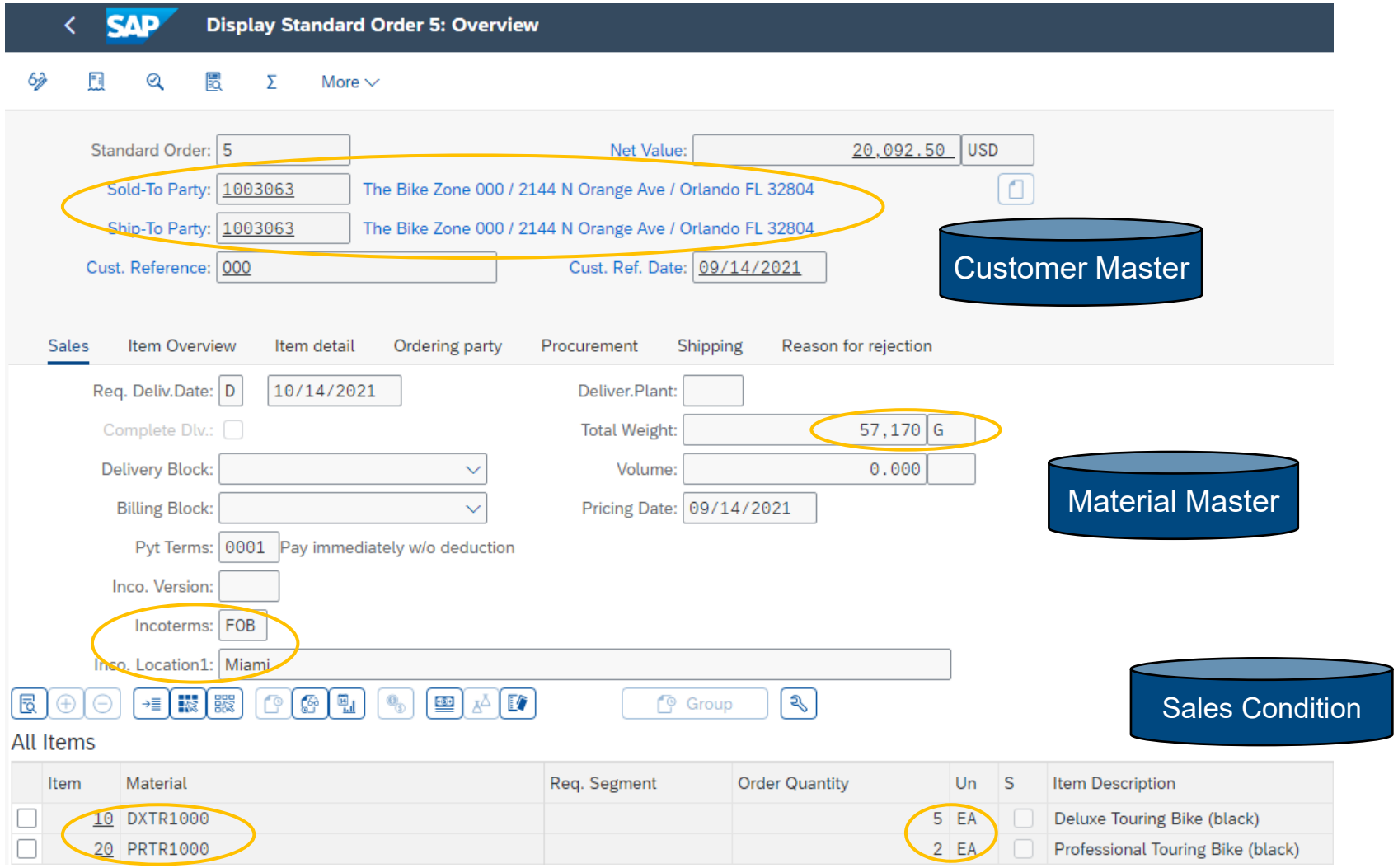
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Unit Overview

- SD Organizational Structure
- SD Master Data
- Order-to-Cash Process

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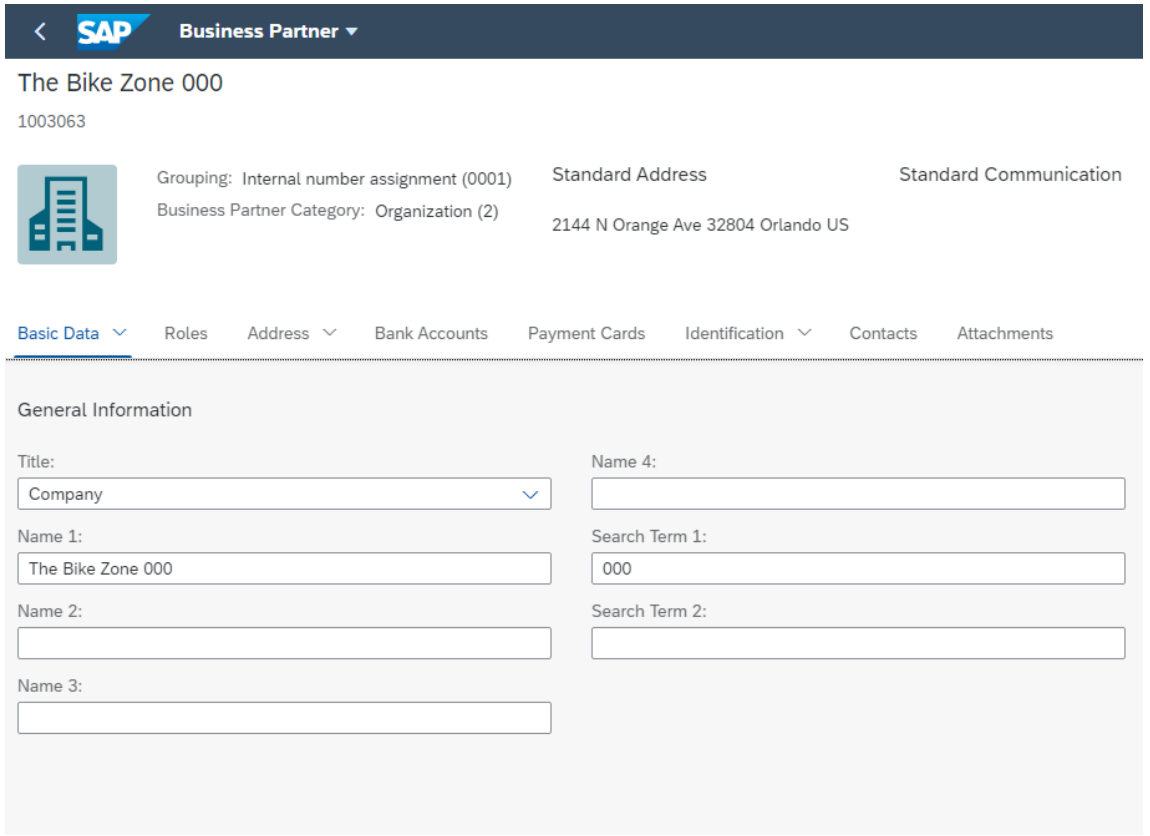
SD Master Data



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Customer Master Data

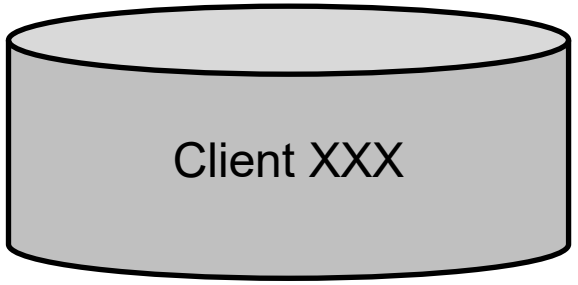
- Customer Master
  - Contains all of the information necessary for processing orders, deliveries, invoices and customer payment
  - Every customer MUST have a master record
    - But you can have one-time customers
- Created by Sales Area
  - Sales Organization
  - Distribution Channel
  - Division
- The customer master information is divided into 3 areas:
  - General Data
  - Company Code Data
  - Sales Area Data



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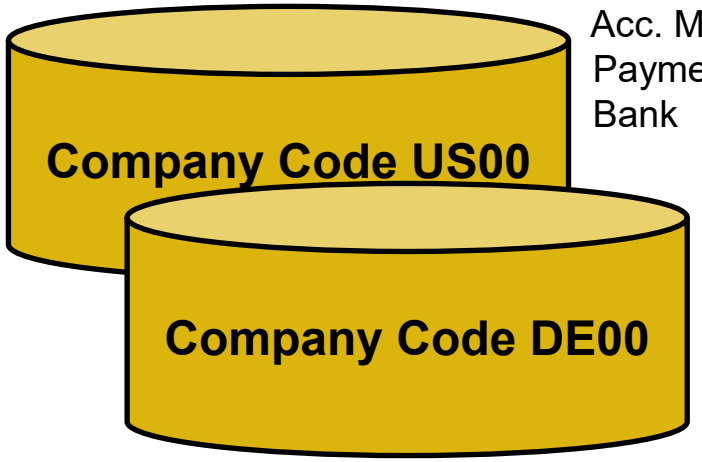
Customer Master Data

General Information relevant for the entire organization:



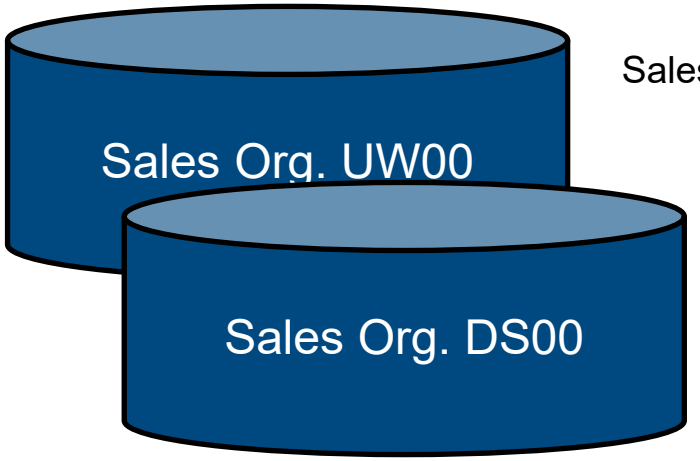
Name
Address
Communication

Company Code specific information:



Acc. Mgmt
Payment
Bank

Sales Area specific information:



Sales Office Currency

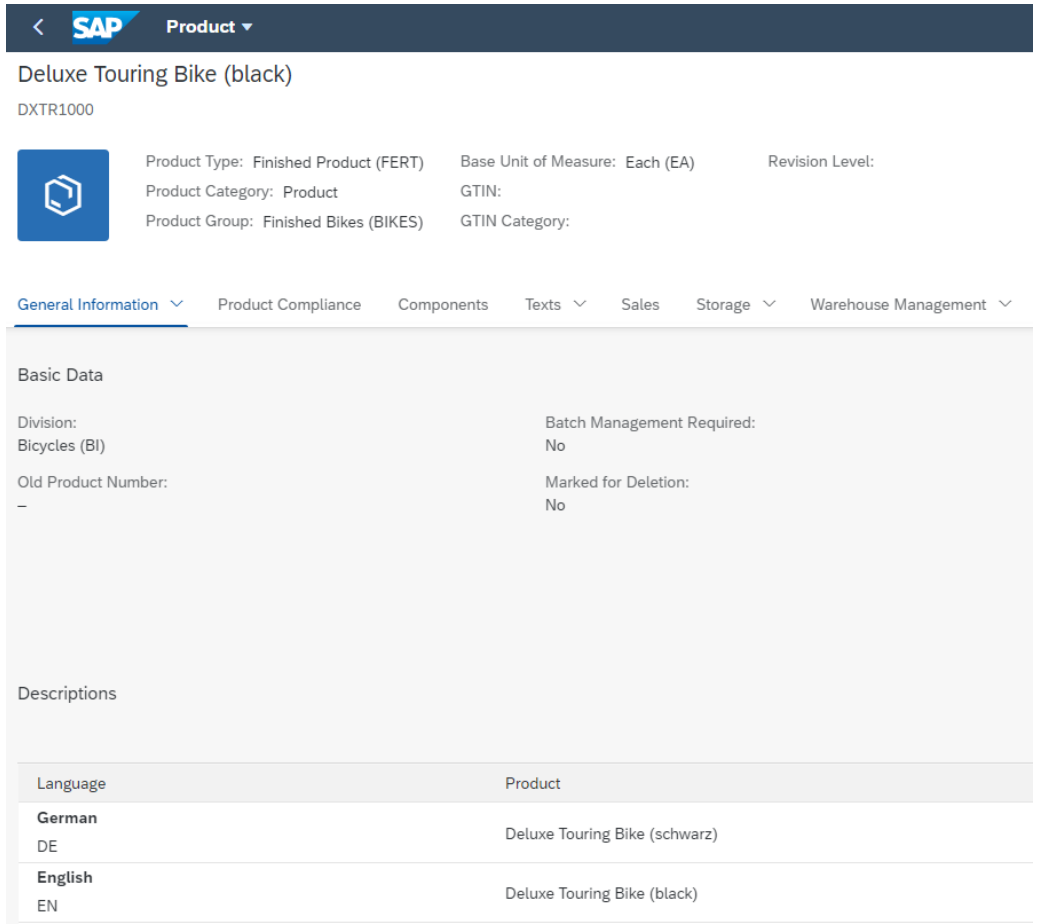
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Material Master Data

- Material Master contains all the information a company needs to manage about a material

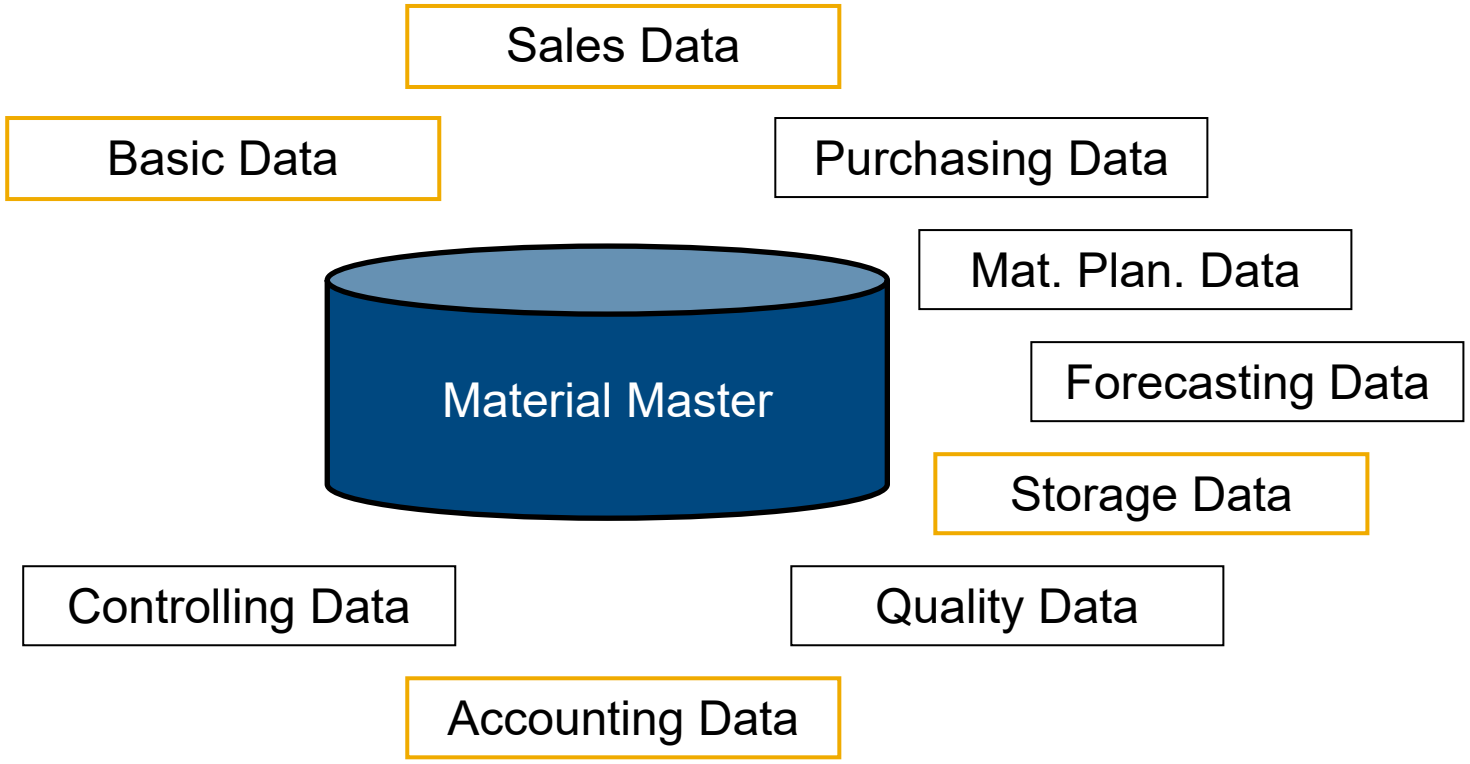
- It is used by most components within the SAP system
  - Sales and Distribution
  - Materials Management
  - Production
  - Plant Maintenance
  - Accounting/Controlling
  - Quality Management

- Material master data is stored in functional segments called Views



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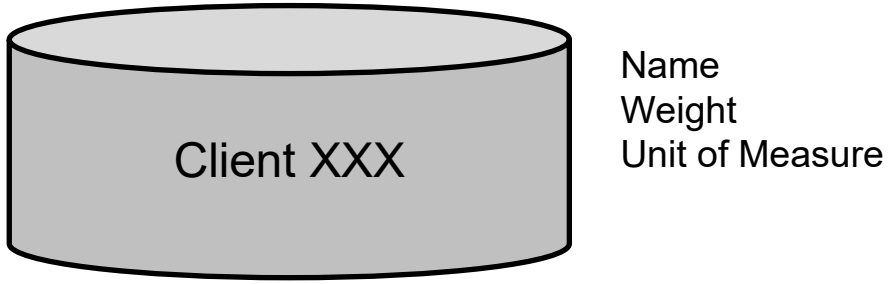
Material Master Views



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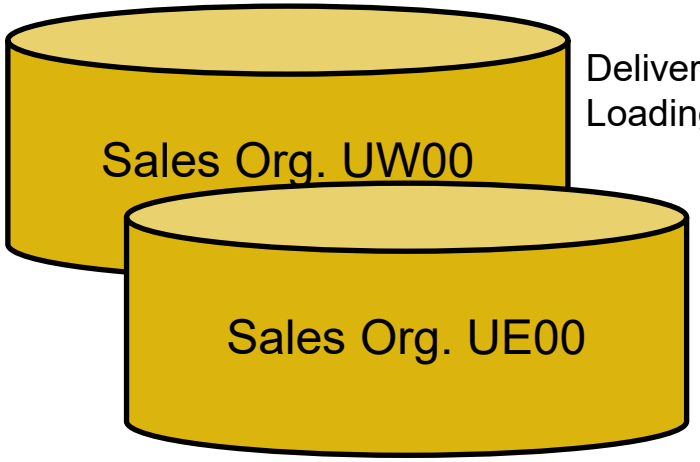
Material Master

General Information relevant for the entire organization:



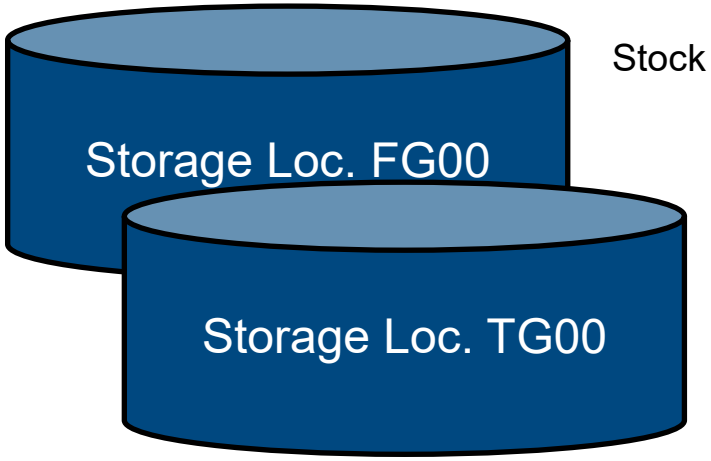
Name
Weight
Unit of Measure

Sales specific information:



Delivering Plant
Loading Grp

Storage Location specific information:

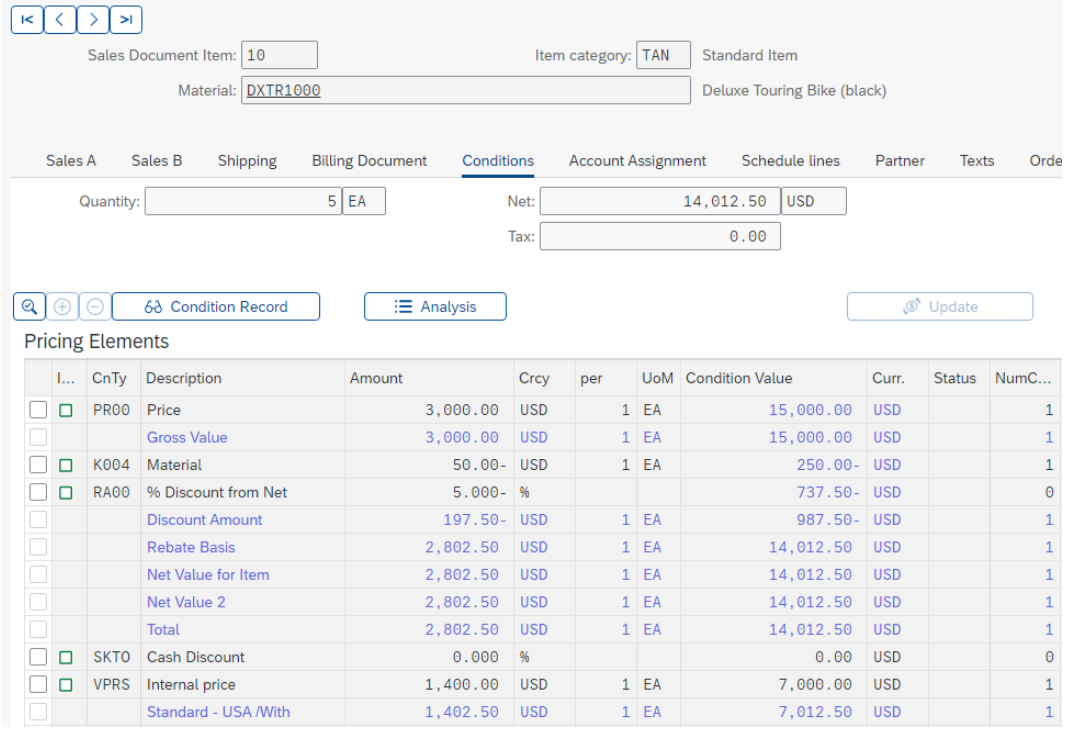


Stock Qty

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Condition Master Data (Pricing)

- Condition master data includes:
  - Prices
  - Surcharges
  - Discounts
  - Freights
  - Taxes
- You can define the condition master to be dependent on various data:
  - Material specific
  - Customer specific
- Conditions can be dependent on any document field



I...	CnTy	Description	Amount	Crcy	per	UoM	Condition Value	Curr.	Status	NumC...
	PR00	Price	3,000.00	USD	1	EA	15,000.00	USD		1
		Gross Value	3,000.00	USD	1	EA	15,000.00	USD		1
	K004	Material	50.00-	USD	1	EA	250.00-	USD		1
	RA00	% Discount from Net	5.000-	%			737.50-	USD		0
		Discount Amount	197.50-	USD	1	EA	987.50-	USD		1
		Rebate Basis	2,802.50	USD	1	EA	14,012.50	USD		1
		Net Value for Item	2,802.50	USD	1	EA	14,012.50	USD		1
		Net Value 2	2,802.50	USD	1	EA	14,012.50	USD		1
		Total	2,802.50	USD	1	EA	14,012.50	USD		1
	SKTO	Cash Discount	0.000	%			0.00	USD		0
	VPRS	Internal price	1,400.00	USD	1	EA	7,000.00	USD		1
		Standard - USA /With	1,402.50	USD	1	EA	7,012.50	USD		1

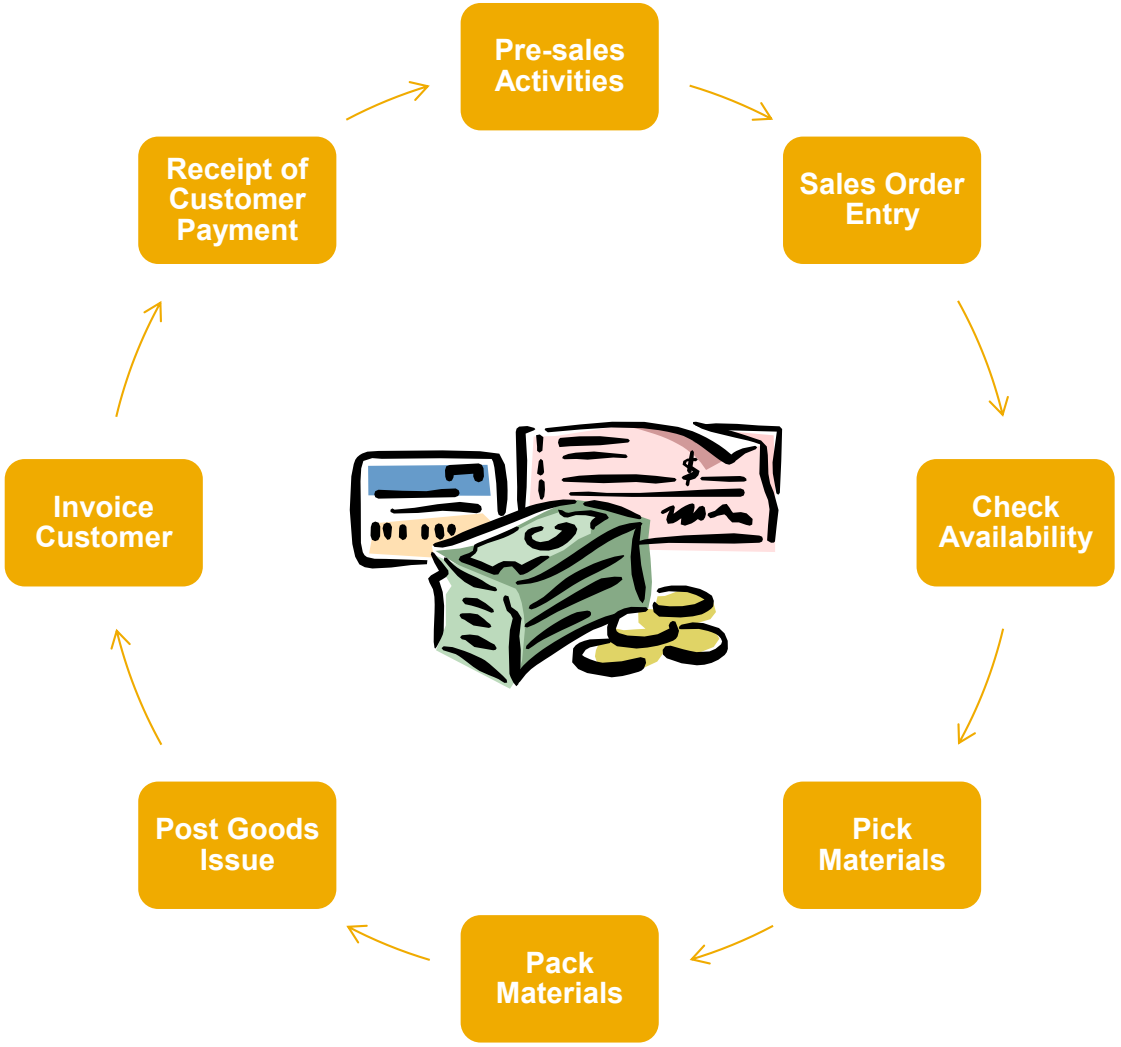
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Unit Overview

- SD Organizational Structure
- SD Master Data
- Order-to-Cash Process

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Sales Order Process



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Pre-Sales Activities (CRM Light)



- **Sales Support** is a component of SD that assists the processes of sales, distribution, and marketing of companies as well as their product and services to its customers

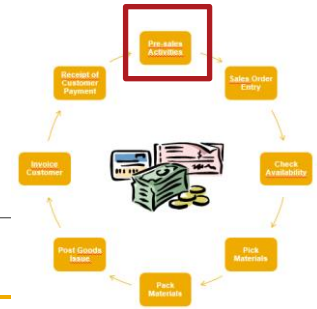
- It contains the following functionality
  - Creating and tracking customer contacts and communications (sales activity)
  - Implementing and tracking direct mailing, internet, and trade fair campaigns based on customer attributes

- Pre-sales documents: Inquiries and Quotations

- The ultimate goal of all pre-sales activities is to equip the sales technician with all the **information** necessary to negotiate and complete the potential sale
  - Past sales activity
  - Past communication
  - Contact information
  - General company info
  - Credit limits and usage
  - Current backorders

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Inquiry



- An inquiry is a **customer's request to a company for information or quotation in respect to their products or services** without any obligation to purchase
  - How much will it cost
  - Material/Service availability
  - May contain specific quantities and dates
- The inquiry is maintained in the system and a quotation is created to address questions for the potential customer.

Company
The Bike Zone
2105 N Lewis Ave
Portland OR 97227

Inquiry

Number/Date
10000002 / 04/21/2010
Reference no./Date
000 /04/21/2010
Delivery date
Day 04/21/2010
Cust. no.
1301
Validity period
04/21/2010 bis 05/21/2010

We deliver according to the following conditions: Currency USD
Terms of payment Payable immediately without deduction

Terms of delivery FOB San Diego

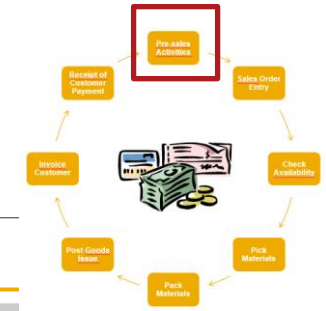
Weights (gross/net) - Volume - Mark
Gross weight 57,170 G Net weight 57,170 G

Please see our promotional offer enclosed. Delivery as long as stocks last.

Item	Material	Qty	Description	Price	Price unit	Value
000010	DXTR1000	5 EA	Deluxe Touring Bike (black)	3,000.00	USD 1 EA	15,000.00
000020	PRTR1000	2 EA	Professional Touring Bike (black)	3,200.00	USD 1 EA	6,400.00
Final amount						21,400.00

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Quotation



- The quotation presents the customer **with a legally binding offer**
  - to deliver specific products or a selection of a certain amount of products
  - in a specified timeframe
  - at a pre-defined price.
- Legally binding on the vendor
- Conditions can apply at various levels:
  - to the entire quotation
  - at item level, to the material to be supplied or to the planned procurement in the case of services
  - at service-line level for individual services (tasks or activities)

Quotation

Company
The Bike Zone
2105 N Lewis Ave
Portland OR 97227

Number/Date
20000000 / 04/21/2010
Reference no./Date
000 /04/21/2010
Delivery date
Day 04/21/2010
Cust. no.
1301
Validity period
04/21/2010 bis 05/21/2010

We deliver according to the following conditions: Currency USD
Terms of payment Payable immediately without deduction

Terms of delivery FOB San Diego

Weights (gross/net) - Volume - Mark
Gross weight 57,170 G Net weight 57,170 G

Please see our promotional offer enclosed. Delivery as long as stocks last.

Item	Material	Description	Qty	Price	Price unit	Value
000010	DXTR1000	Deluxe Touring Bike (black)				
			5 EA	3,000.00	USD 1 EA	15,000.00
	Material			50.00-	USD 1 EA	250.00-
	% Discount from N			5.000-	%	737.50-
	Net Value for Ite			2,802.50	USD 1 EA	14,012.50
000020	PRTR1000	Professional Touring Bike (black)				
			2 EA	3,200.00	USD 1 EA	6,400.00
	% Discount from N			5.000-	%	320.00-
	Net Value for Ite			3,040.00	USD 1 EA	6,080.00
Final amount						20,092.50

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Sales Order (1)



- Sales order processing can originate from a variety of documents and activities
  - Customer contacts us for an order: phone, internet, e-mail
  - Existing Contract
  - Quotations

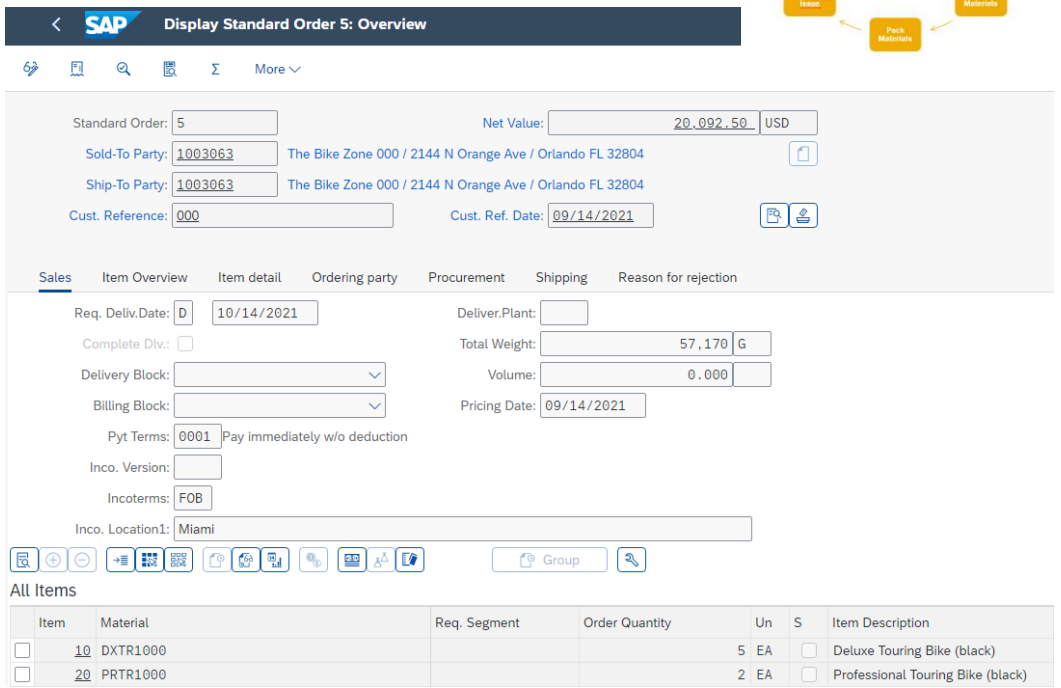
- The electronic document that is created should contain the following basic information:
  - Customer Information
  - Material/service and quantity
  - Pricing (conditions)
  - Specific delivery dates and quantities
  - Shipping information
  - Billing Information

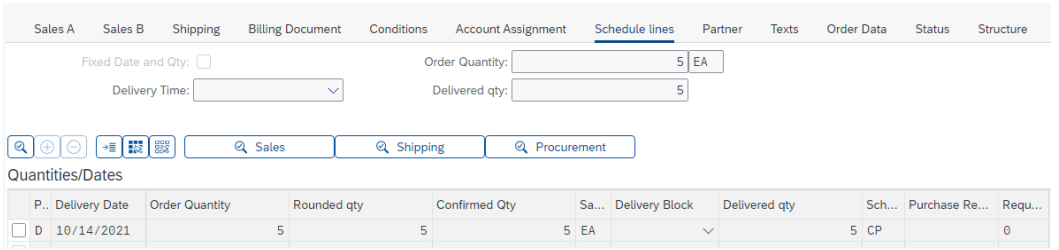
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Sales Order (2)



- The sales document is made up of three primary areas:
  - Header
    - Data relevant for the entire sales order: e.g.: customer data, total cost of the order
  - Line Item(s)
    - Information about the specific product: e.g.: material and quantity, cost of an individual line
  - Schedule Line(s)
    - Uniquely belongs to a Line Item, contains delivery quantities and dates for partial deliveries





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Sales Order (3)



- The sales order contains all of the information **needed to process your customers request**

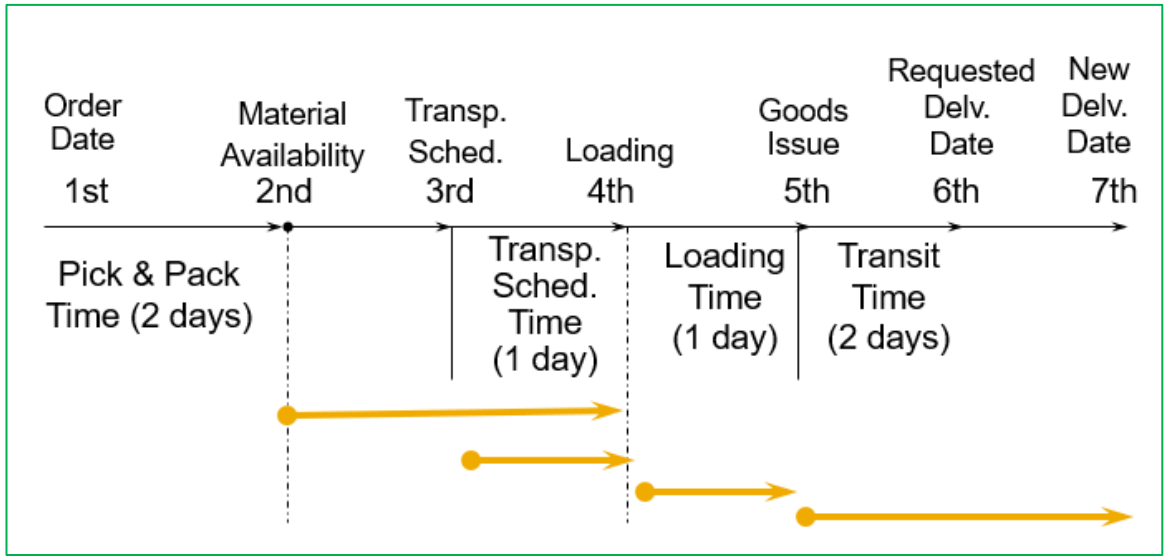
- The following information is determined for each sales order:
  - Delivering Schedule
  - Shipping point and route determination
  - Availability Check
  - Transfer of requirements to MRP
  - Pricing
  - Credit limit check

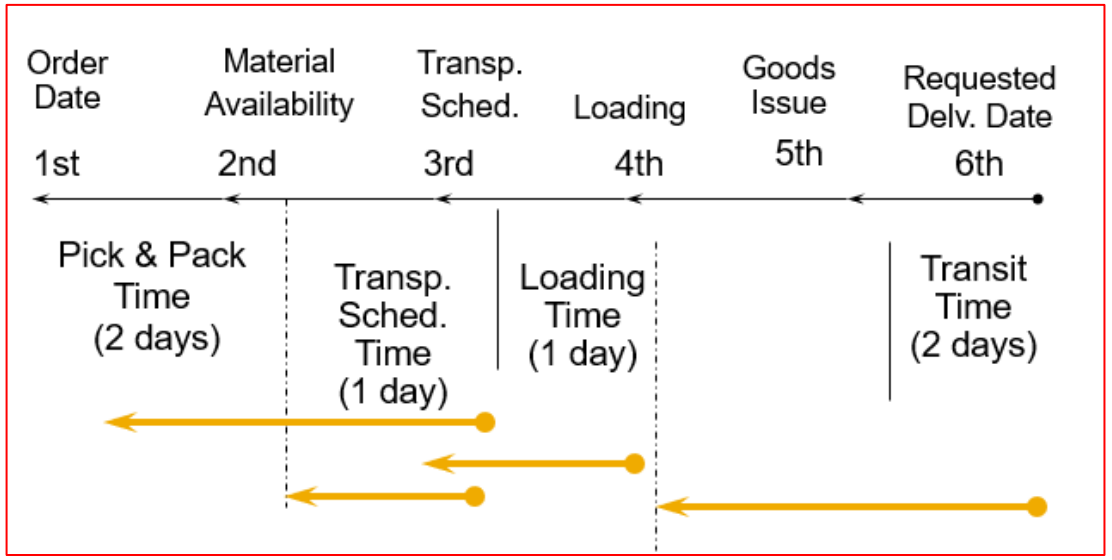
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Sales Order: Delivery Scheduling



- When an order is created you **must enter a requested delivery date** for the order or each line item.
- The system **will then determine a delivery timeline**, this will be used when determining our material availability, or ATP (Availability to Promise) date.
- The system will determine this date using **forward** **and** **backward** scheduling rules you have defined.





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Sales Order: Shipping & Route Determination



- During the creation of the sales order, the system must **determine the shipping point from which the material will be shipped and the route the material will take** to get from your warehouse to your customers location.
- A shipping point is determined for **each line item within the order**.
- The route determination is used to **define the transit time of the material** that we used in scheduling.

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Sales Order: Pricing



- The system displays pricing information for all sales documents on the pricing screens at both the header and the line item level
  - **Header pricing is valid for the whole order** and it is the cumulative of all line items within the order
  - **Line item pricing is for each specific material**.
- The system **will automatically search for price, discounts, surcharges, calculated taxes and freight**
- You have the ability to **manually manipulate the pricing at both the header and line item** level within the sales order by entering a condition type and amount
  - Taxes and freight can be set-up automatically so we can not manually enter them

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Credit Check



- Allows your company to manage its credit exposure and risk for each customer by specifying credit limits
- During the sales order process the system will alert the sales report about the customers' credit situation that arises
- If necessary, the system can be configured to block orders and deliveries

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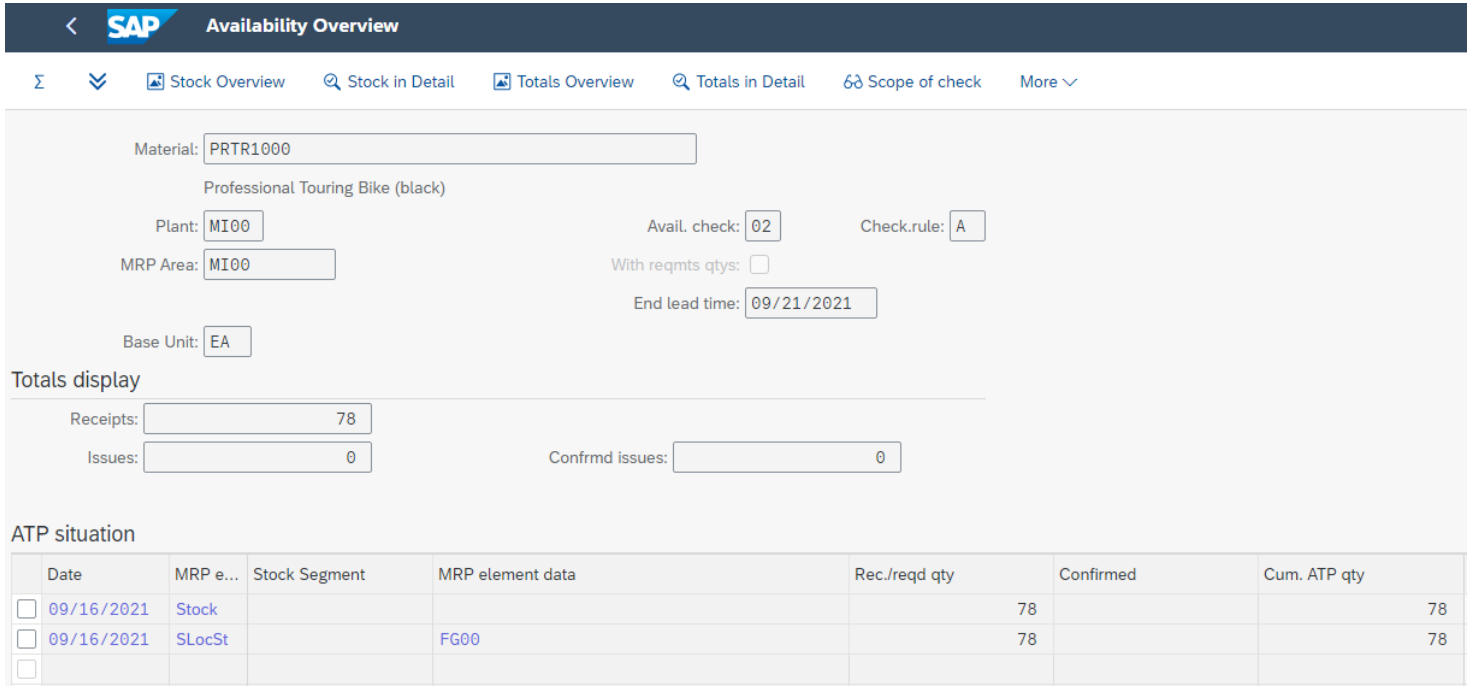
Availability Check



- Availability Check
  - Determines the material availability date
  - **Considers all inward and outward inventory** movements

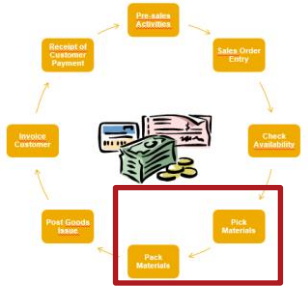
- Proposes 3 methods of delivery:
  - One-time delivery
  - Complete delivery
  - Delayed proposal

- Rules are created by USERS



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Shipping & Transportation

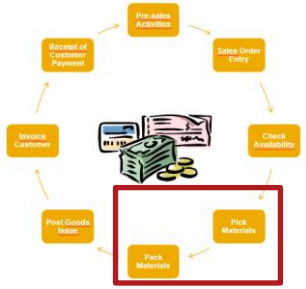


- The shipping process begins when **you create the delivery document for the sales order**. This document controls, supports, and monitors numerous sub-processes for progressing the shipping:
  - Picking
  - Packing
  - Post Goods Issue

- Integrated with the Material Management (MM), extended Warehouse Management (eWM) and Finance (FI) modules

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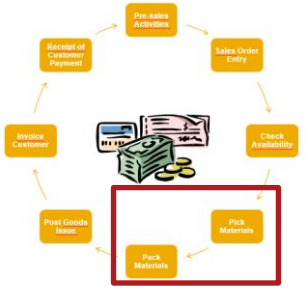
Delivery Creation



- Checks order and materials to determine if a delivery is possible — delivery block (hold), completeness
- Confirms availability
- Determines total weight & volume
- Generates packing proposal
- Checks scheduling
- Considers partial deliveries
- Updates route assignment
- Assigns picking location
- Updates sales order
- Determines batches
- Quality check (if needed)
- Updates sales order

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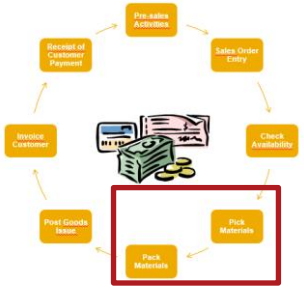
Picking

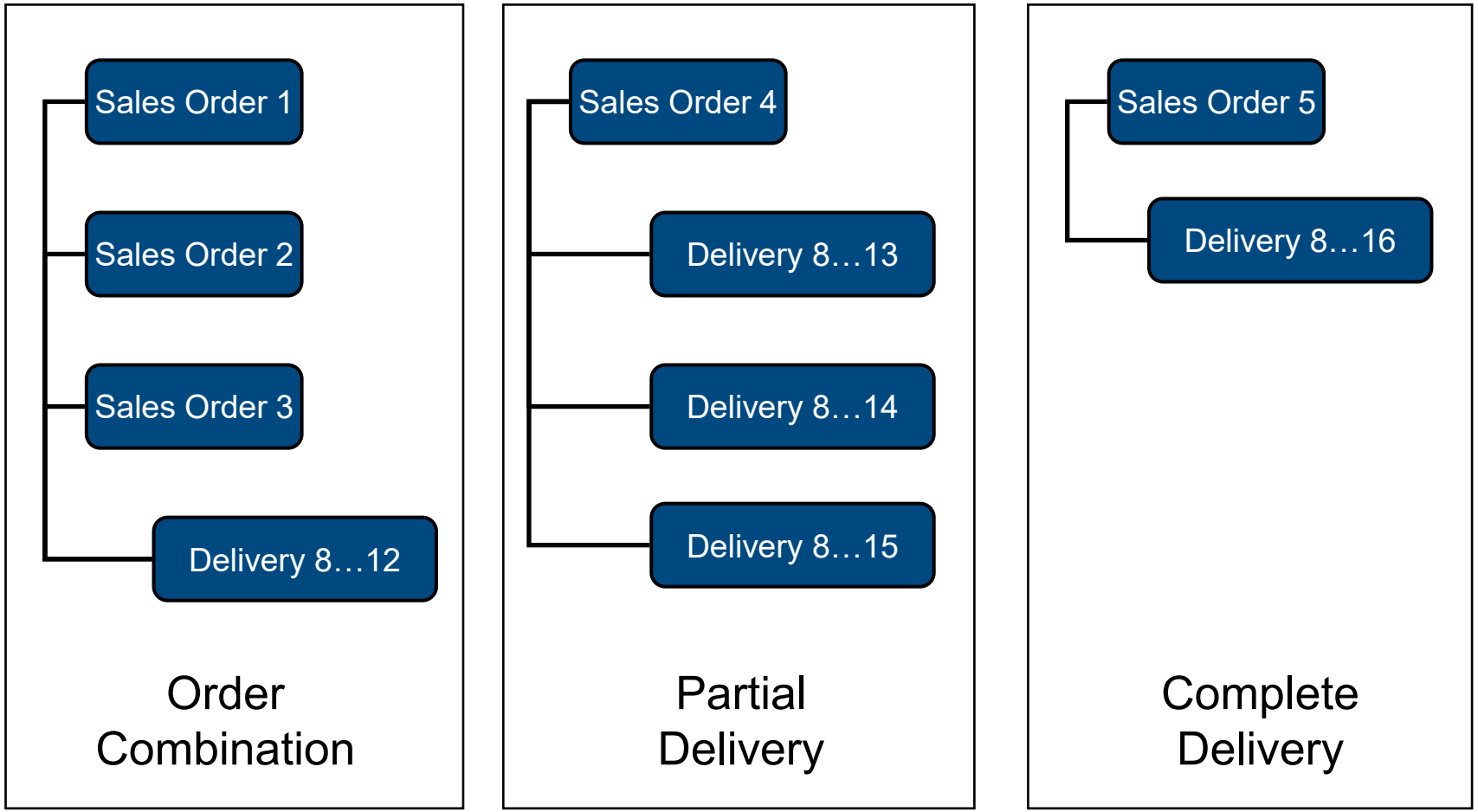


- Quantities based on delivery note
- Assigned date when picking should begin
- Automated storage location assignment
- Supports serial number/lot number tracking and batch management
- Integrated with Warehouse Management (WM)

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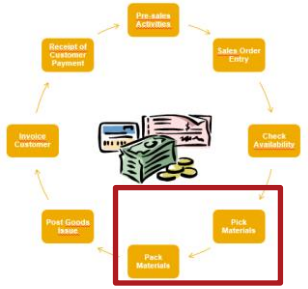
Shipping





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Loading and Packing



- Identifies **which packaging is to be used for specified products** (customer preference and UCC-128 considerations)
- Identifies and updates accounts associated with returnable packaging
- Tracks the packed product by container
- Insures weight/volume restrictions are enforced
- All packed items are assigned to the required means of transportation

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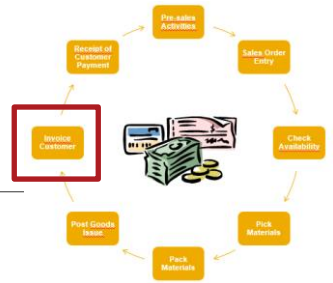
Goods issue



- Event **that indicates the legal change in ownership of the products**
  - It signals that the goods have left the premises of the seller. Legal change in ownership depends on the “Incoterm” and what is written in the sales order terms and conditions.
- **Reduces inventory and enters Cost of Goods Sold**
- Automatically **updates the General Ledger (G/L) accounts**
- Ends the shipping process and updates the status of the shipping documents

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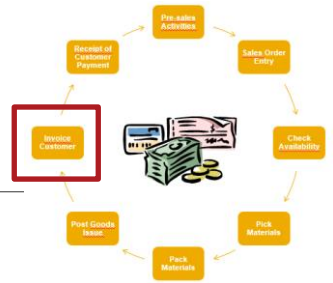
Billing



- The billing document is created by **coping data from the sales order and/or delivery document**.
  - Order-based billing
  - Delivery-based billing
- The billing process is used to **generate the customer invoice**
- It will update the customer's credit status.
- The billing document will **automatically create a debit posting to your customer's sub-ledger account** and credit it to your revenue account
- It is at this point that the sales process is passed over to Financial Accounting to await payment.

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Billing Methods



Delivery based Invoicing

- Order 6
  - Delivery 8…20 → Invoice 9…45
  - Delivery 8…21 → Invoice 9…46

Collective Invoicing

- Order 9
  - Delivery 8…33
  - Delivery 8…34
- Order 14
  - Delivery 8…56

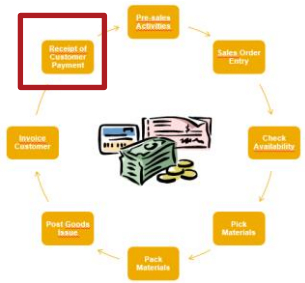
Delivery 8…33, Delivery 8…34, Delivery 8…56 → Invoice 9…68

Split Invoicing

- Order 32
  - Delivery 8…86 → Invoice 9…91
  - Delivery 8…86 → Invoice 9…92

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Payment



- Payment is the final step in the sales order process, this step is managed by the Financial Accounting department.

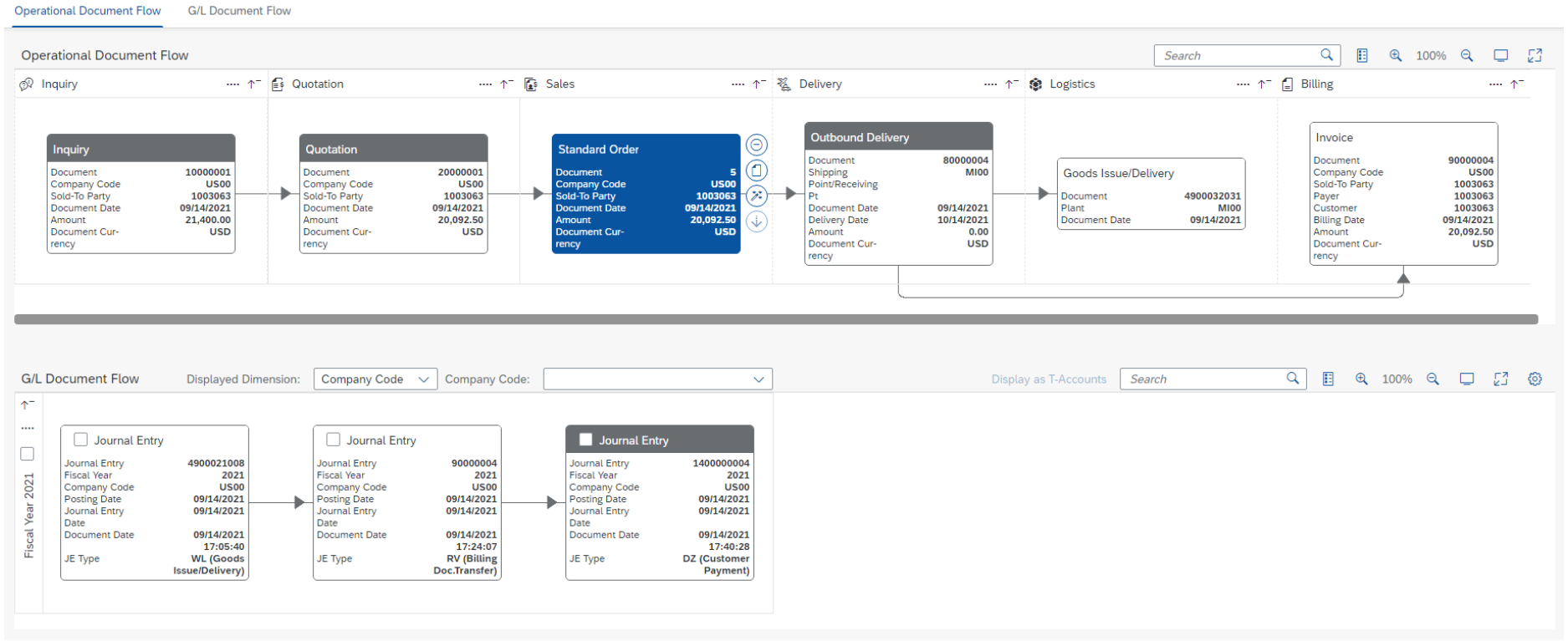
- Final payment includes:
  - **Posting payments against invoices**
  - **Reconciling differences between payment and invoice**

- Payment will **create a posting clearing the liability in the A/R account and increases your bank account**.

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Document Flow

- The document flow and order status feature allows you to find the status of an order at any point in time. The SAP updates the order status every time a change is made to any document created in the customer order management cycle (Order-to-Cash).



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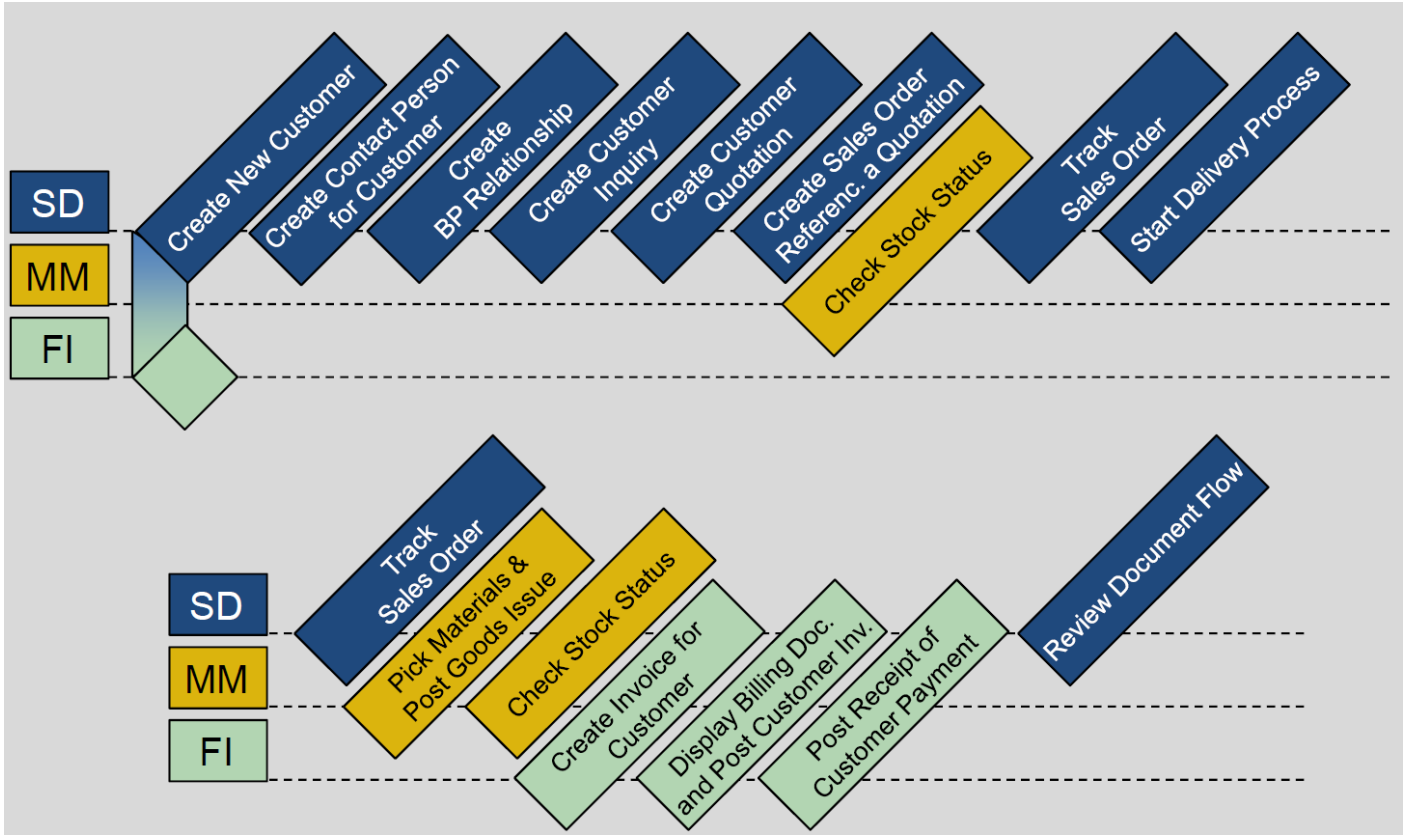
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Agenda

- Purchase-to-pay vs. Order-to-cash processes
- SD Organizational Structure
- SD Master Data
- Order-to-Cash Process
- Outlook SD Case Study

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Outlook: SD Case Study



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Learning Objectives (What?)

Students can…

- … describe the **general order-to-cash process on the basis of the essential activities** and **differentiate it from an procurement process**.
- … describe the **organizational structures and master data** required for Sales and Distribution with SAP SD.
- … create **customer master data relevant for sales** with SAP SD.
- … **explain the integrated order-to-cash process and execute** it in SAP-SD